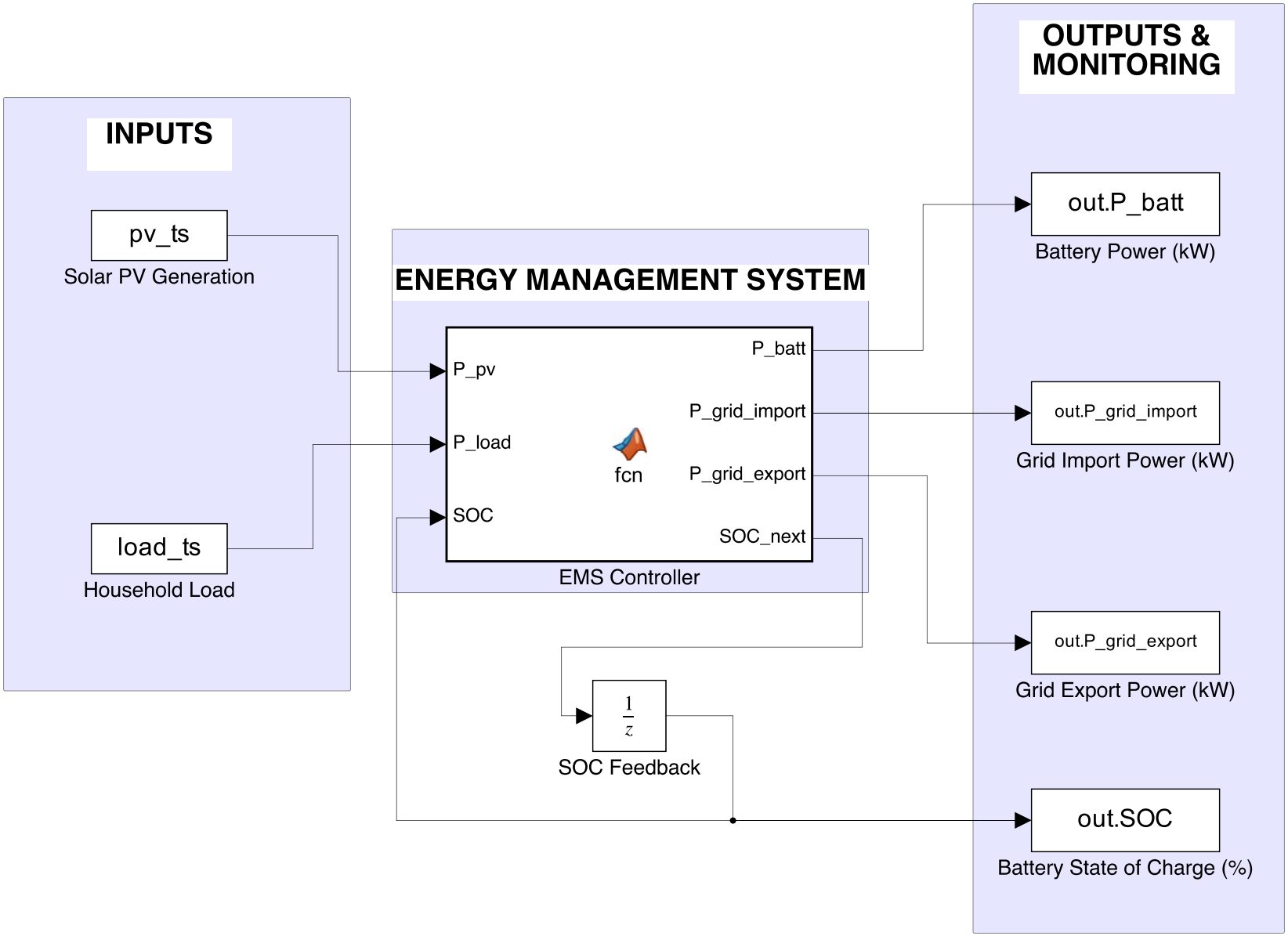

INPUTS
pv_ts
Solar PV Generation
load_ts
Household Load
ENERGY MANAGEMENT SYSTEM
P_pv
P_load
SOC
fcn
P_batt
P_grid_import
P_grid_export
SOC_next
EMS Controller
1/z
SOC Feedback
OUTPUTS & MONITORING
out.P_batt
Battery Power (kW)
out.P_grid_import
Grid Import Power (kW)
out.P_grid_export
Grid Export Power (kW)
out.SOC
Battery State of Charge (%)